

Shop Inventory and Sales Management System

Project Documentation

Cover Page

Shop Inventory and Sales Management System

Professional Business Management Solution for Retail and Merkato-Style Businesses

Prepared By: Michael Bahiru

Project Type: Final GC year internship / Business Information Systems Project

Technology Stack: Python, Flask, MySQL, HTML, CSS, JavaScript, Bootstrap

Database: MySQL

Development Environment: VS Code

Project Version: Version 2.0

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Acknowledgment

First and foremost, I would like to thank **Almighty God** for giving me the strength, patience, and opportunity to complete this project successfully.

I would like to express my sincere gratitude to my instructors, advisor, classmates, friends, and family members for their continuous encouragement, guidance, and support throughout the development of this project.

I would especially like to thank my friend and classmate, Addis Berhanu, who invited me to work and observe real business operations in Merkato(Addis Mercato). That experience gave me practical exposure to inventory handling, supplier-credit relationships, price fluctuations, and the operational challenges faced by real businesses. The insights gained from observing Merkato businesses played a major role in shaping the logic and architecture of this system, particularly the dynamic pricing and payable management components.

I also extend my appreciation to everyone who provided valuable ideas, suggestions, technical support, and business insights related to inventory management, sales handling, receivable and payable systems, and real-world retail business operations.

Finally, I would like to thank my family for their patience, motivation, and moral support throughout this journey.

Abstract

The Shop Inventory and Sales Management System is a web-based business management application designed to help shops, retail businesses, wholesalers, and Merkato - style trading businesses manage their daily operations efficiently.

The system automates important business processes such as inventory management, purchasing, sales handling, customer receivables, supplier payables, reporting, and staff authentication.

One of the major strengths of the system is its practical adaptation to inflation-sensitive businesses such as Merkato markets, where product prices frequently change due to import costs, currency fluctuations, transportation expenses, and international market conditions.

To solve this real-world problem, the system uses a dynamic pricing architecture where products do not store permanent selling prices or buying prices. Instead, each purchase transaction records its own buying price, extra costs, and selling price. This allows the system to:

- Track historical purchase prices
- Support inflation-based price variation
- Preserve accurate cost calculations
- Calculate realistic profits
- Support business decision-making

The system also includes role-based authentication where:

- Admin users access the full dashboard
- Sales staff access only sales operations
- Purchasers access inventory and purchasing operations

The project was developed using Python Flask framework with MySQL database and responsive Bootstrap user interfaces.

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1. Introduction

Business management systems are essential tools for modern retail and wholesale operations. Manual business processes often create problems such as inaccurate stock tracking, lost transaction records, poor debt management, and incorrect profit calculations.

This project was developed to digitize and automate those processes using a professional web-based system.

The Shop Inventory and Sales Management System is designed specifically to support businesses operating in dynamic market environments such as Merkato, where product prices frequently fluctuate.

The system helps business owners:

- Track inventory accurately
- Manage purchases and suppliers
- Record sales professionally
- Handle customer debts and supplier debts
- Monitor profits and revenues
- Generate business reports
- Manage staff roles securely

2. Problem Statement

Many small and medium businesses still use manual notebooks or basic spreadsheets for managing inventory and sales.

This creates several problems:

- Incorrect stock quantities
- Difficulty tracking supplier debts
- Difficulty tracking customer receivables
- Poor profit calculation
- Lack of historical purchase tracking
- Weak reporting system
- No role-based staff management
- Data inconsistency during inflation

In Merkato-style businesses, prices frequently change because many products are imported. Traditional systems that store one fixed buying price and one fixed selling price for products fail to represent the real business situation.

Therefore, there is a need for a dynamic and professional business management system.

3. Objectives

General Objective

To develop a professional web-based Shop Inventory and Sales Management System that automates business operations.

Specific Objectives

- To manage product inventory effectively
- To record purchases dynamically
- To manage sales transactions
- To support credit purchases and credit sales
- To manage supplier payables
- To manage customer receivables
- To calculate business profit accurately
- To generate business reports
- To provide role-based access control
- To create mobile-friendly business interfaces

4. Scope of the System

The system includes the following functionalities:

- Product and category management
- Supplier management
- Customer management
- Purchase registration
- Sales registration
- Inventory tracking
- Payable management
- Receivable management
- Financial reporting
- User authentication and authorization
- Role-based access control

The system is intended for small and medium retail businesses.

5. Existing System Analysis

Most traditional businesses rely on:

- Manual bookkeeping
- Excel sheets
- Paper receipts
- Human memory

Limitations include:

- Time-consuming operations
- Human error
- Data loss risk
- Difficult report generation
- Poor stock visibility
- Weak debt tracking

The proposed system solves these issues through automation.

6. Proposed System

The proposed system is a web-based application developed using Flask and MySQL.

The system introduces:

- Centralized database management
- Automated calculations
- Real-time stock updates
- Dynamic pricing architecture
- Historical transaction recording
- Role-based login system
- Mobile-friendly interface

The system is optimized for businesses affected by inflation and fluctuating market prices.

7. System Features

Core Features

Inventory Management

- Add product categories
- Add products
- Edit products
- Purchase-based stock increment
- Dynamic latest price display

Purchase Management

- Supplier selection
- Buying price registration
- Extra cost recording
- Selling price registration
- Credit purchase handling
- Purchase history tracking

Sales Management

- Real-time stock deduction
- Cart system
- Dynamic selling price usage
- Credit sales support
- Receipt generation

Receivable Management

- Customer debt tracking
- Payment registration
- Payment history ledger
- Balance management

Payable Management

- Supplier debt tracking
- Partial payment support
- Ledger-style payment history
- Overpayment validation

Reporting System

- Revenue reports
- Profit reports

- Sales trend visualization
- Top-selling products
- Date filtering

Authentication System

- Login system
- Password hashing
- Role-based authorization
- Staff registration

8. Technologies Used

Backend Technologies

Python

Used as the primary programming language.

Flask

Used as the web framework for routing and backend logic.

MySQL

Used for database management.

Frontend Technologies

HTML

Used for page structure.

CSS

Used for styling.

Bootstrap

Used for responsive and mobile-friendly design.

JavaScript

Used for interactive features.

Security Technologies

Werkzeug Password Hashing

Used for secure password storage.

9. Database Design

Main Tables

users

Stores system staff information.

categories

Stores product categories.

products

Stores product names and categories.

suppliers

Stores supplier information.

customers

Stores customer information.

purchases

Stores purchase transactions.

purchase_items

Stores detailed purchased products including:

- buying price
- extra cost
- selling price
- quantity

sales

Stores sales transactions.

sale_items

Stores sold products including:

- unit selling price
- cost price
- quantity

payables

Stores supplier debts.

payable_payments

Stores supplier payment history.

receivables

Stores customer debts.

receivable_payments

Stores customer payment history.

10. System Modules

Inventory Module

Handles stock visibility and product management.

Purchase Module

Handles stock purchasing and supplier management.

Sales Module

Handles customer sales operations.

Receivable Module

Handles customer debts.

Payable Module

Handles supplier debts.

Reporting Module

Handles business analytics.

Authentication Module

Handles user login and authorization.

11. Authentication and Authorization

The system uses role-based authentication.

Roles

Admin

Can access all modules.

Sales

Can access sales-related pages only.

Purchaser

Can access inventory and purchasing pages.

This improves:

- Security
- Staff accountability
- Operational control

12. Inventory Management Logic

The inventory system was redesigned professionally.

Products now store only:

- Product name
- Product category

The system no longer stores fixed buying prices or selling prices in the product table.

This design supports:

- Dynamic pricing
- Inflation handling
- Historical price tracking
- Realistic profit calculation

The latest purchase determines the latest selling price shown in inventory.

13. Purchase Management Logic

Purchase registration includes:

- Supplier
- Buying price
- Extra cost
- Quantity
- Selling price
- Credit status
- Description

The system calculates:

$\text{Unit Cost} = \text{Buying Price} + \text{Extra Cost}$

$\text{Total Amount} = \text{Unit Cost} \times \text{Quantity}$

If the purchase is on credit, the payable amount only includes:

$\text{Buying Price} \times \text{Quantity}$

Extra costs are excluded because they are self-paid operational costs.

This reflects real Merkato business practices.

14. Sales Management Logic

The sales system records:

- Selling price
- Quantity
- Cost price at sale time

This preserves historical cost accuracy.

The system supports:

- Dynamic selling prices
- Inflation-aware profit tracking
- Stock deduction
- Credit sales
- Receipt generation

15. Receivable Management

The receivable system tracks:

- Customer debts
- Partial payments
- Payment references
- Remaining balances

The system includes:

- Ledger-style payment history
- Mobile-friendly interface
- Validation logic

16. Payable Management

The payable system tracks:

- Supplier debts
- Credit purchases
- Partial payments
- Payment history

Features include:

- Over-payment prevention
- Ledger accounting view
- Supplier dashboard
- Remaining balance calculation

17. Reporting System

The reporting system provides:

- Daily sales reports
- Weekly sales reports
- Monthly sales reports
- Revenue analysis
- Net profit analysis
- Sales trend charts
- Top-selling products

Profit calculation uses historical cost prices stored during sales.

This ensures accurate financial reporting.

18. User Interface Design

The system uses Bootstrap responsive design.

Features include:

- Mobile-friendly layouts
- Responsive tables
- Dashboard cards
- Clean navigation
- Professional forms
- Ledger-style pages

The design supports both desktop and smartphone usage.

19. Challenges and Solutions

Challenge 1: Inflation-Based Price Variation

Problem

Product prices frequently change.

Solution

Move buying price and selling price from products table to purchase_items table.

Challenge 2: Accurate Profit Calculation

Problem

Changing prices affect profit accuracy.

Solution

Store cost_price inside sale_items during every sale.

Challenge 3: Credit Management

Problem

Businesses need debt tracking.

Solution

Develop receivable and payable ledger systems.

Challenge 4: Staff Access Control

Problem

Different employees need different permissions.

Solution

Implement role-based authorization.

20. Future Enhancements

Future improvements may include:

- Barcode scanning
- QR receipt system
- Multi-branch support
- Cloud deployment
- SMS notifications
- Mobile application
- AI-powered business forecasting
- Expense management
- Tax calculation
- Accounting integration
- Backup and restore system
- Multi-language support

21. Conclusion

The Shop Inventory and Sales Management System successfully automates major business operations for retail and Merkato-style businesses.

The project demonstrates how modern software solutions can solve real-world business challenges such as:

- Inventory management
- Inflation-sensitive pricing
- Supplier debt tracking
- Customer debt tracking
- Profit calculation
- Reporting and analytics

The system is scalable, practical, mobile-friendly, and designed with real business workflows in mind.

This project also serves as a strong foundation for future enterprise-level business management systems.

Author Information

Developed By

Michael Bahiru

Project Focus

- Business Information Systems
- Retail Automation
- Inventory Management
- Financial Tracking Systems
- Business Process Digitalization

Vision

To build scalable business technologies and modern digital systems that improve operational efficiency for businesses and industries.

Contact Information

- Email: michaelbahru7@gmail.com
- LinkedIn:
https://www.linkedin.com/in/%E1%88%9A%E1%8A%AB%E1%8A%A4%E1%88%8D-%E1%89%A3%E1%88%85%E1%88%A9-092592405?utm_source=share_via&utm_content=profile&utm_medium=member_android
- Instagram:
https://www.instagram.com/michael_bahru?igsh=enRqN2xpYXowbHN3
- GitHub: <https://github.com/michaelbahiru>
- Phone: +251 9 02354723

End of Documentation